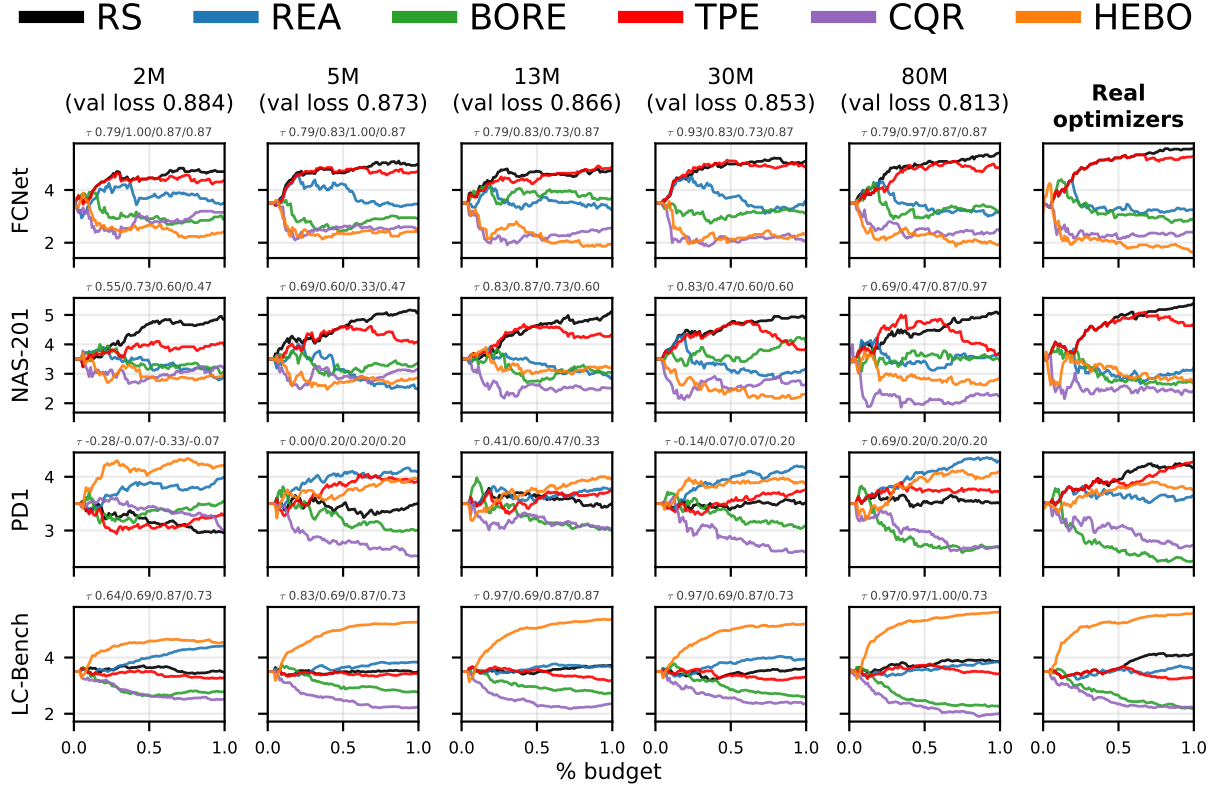
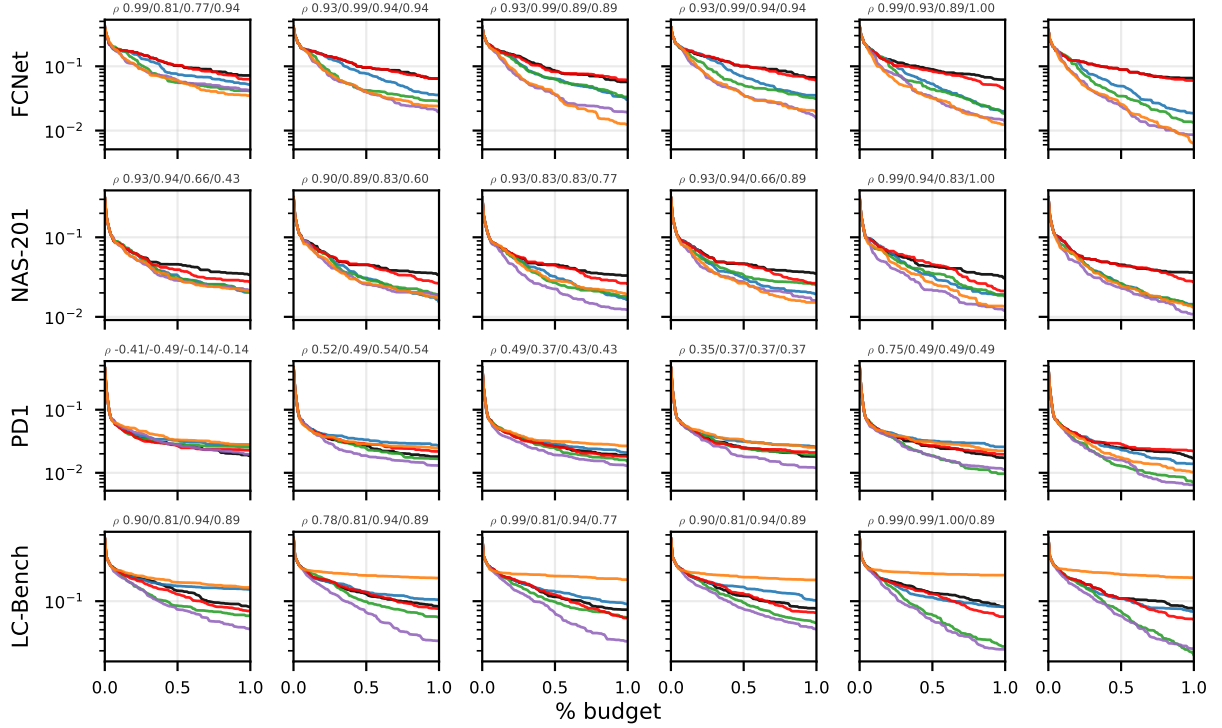




(a) Average ranks.



(b) Average normalized regrets.

Optimization quality vs. model size at a fixed 2B-token budget. Average rank (a) and normalized regret (b) of the learned optimizers (columns: 2M/5M/13M/30M/80M parameters) against the real optimizers (last column), on four validation benchmarks (30 seeds each). We report in the column title the validation loss of each model (lower is better), from 0.884 (2M) to 0.813 (80M). And, on top of each plot, we highlight the correlation between the learned and real optimizers at 25/50/75/100% optimization budget.